Lecture 16
Discrete and Continuous Probability Distributions

Review

1 What are independent events?

Events where one occurring does not change the probability of the other.

2 What is a random variable?

The outcome of a random trial recorded as a number.

3 What is a discrete random variable?

One whose possible values can be listed, like the number of heads in 3 tosses.

4 What is a probability distribution?

A function giving each value of X a probability between 0 and 1; they add to 1.

The mean of a random variable

expected value μ

the long-run average of X: $\mu = \Sigma\, x \cdot P(x)$

variance σ^2

$\sigma^2 = \Sigma\, (x - \mu)^2 \cdot P(x)$, and the standard deviation is $\sigma = \sqrt{\sigma^2}$

three coin tosses

$\mu = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 1.5$ heads

Example: grade points

A STAT 251 exam grade on the 4-point scale, for a student chosen at random:

x	0	1	2	3	4
grade	F	D	C	B	A
P(x)	0.05	0.10	0.30	0.25	0.30

(a) What is P(better than a C)?

(b) What is the mean grade, μ?

(a) P(X = 3) + P(X = 4) = 0.25 + 0.30 = 0.55

(b) μ = 0(0.05) + 1(0.10) + 2(0.30) + 3(0.25) + 4(0.30) = 2.65

Bernoulli trials

Bernoulli trial

one trial with two outcomes: success (X = 1) with probability p, failure (X = 0) with probability 1 − p

X ~ Bernoulli(p)

its single parameter p is the probability of success

mean and variance

$\mu = p$ and $\sigma^2 = p(1 - p)$

Example: guessing one question

You guess on a multiple-choice question with 4 options, one of them correct.

X = 1 if the guess is right, X = 0 if not; p = 1/4 = 0.25

(a) What is P(X = 0)?

(b) What are μ and σ^2?

(a) $1 - 0.25 = 0.75$

(b) $\mu = 0.25$, $\sigma^2 = 0.25 \times 0.75 = 0.1875$

The binomial setting

a fixed number of trials n

the trial is repeated a set number of times, decided in advance

two outcomes, the same p

each trial is a success or a failure, with the same probability of success p every time

independent trials

the outcome of one trial does not change the probability of success on any other

The binomial distribution

$X \sim \text{Binom}(n, p)$

X = the number of successes in n independent trials, each with probability of success p

$P(X = k)$

$\frac{n!}{(k!(n-k)!)} \cdot p^k \cdot (1-p)^{n-k}$: the first factor counts the ways to arrange k successes

mean and variance

$\mu = np$ and $\sigma^2 = np(1-p)$

Example: counting vs. the formula

Monday: X = the number of heads in 3 tosses of a fair coin, counted straight from the sample space.

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}; X ~ Binom(n = 3, p = 0.5)

(a) From S, what are P(X = 3) and P(X = 2)?
(b) Use the formula to find P(X = 3) and P(X = 2).
(c) What does 3! / (2! 1!) count?

(a) 1/8 (HHH) and 3/8 (HHT, HTH, THH)
(b) $3!/(3!\,0!) \cdot 0.5^3 \cdot 0.5^0 = 1/8$ $3!/(2!\,1!) \cdot 0.5^2 \cdot 0.5^1 = 3/8$: the same
(c) the 3 orders with 2 heads: HHT, HTH, THH

Example: guessing a whole quiz

You guess on all 10 questions of a 4-option multiple-choice quiz.

X ~ Binom(n = 10, p = 0.25)

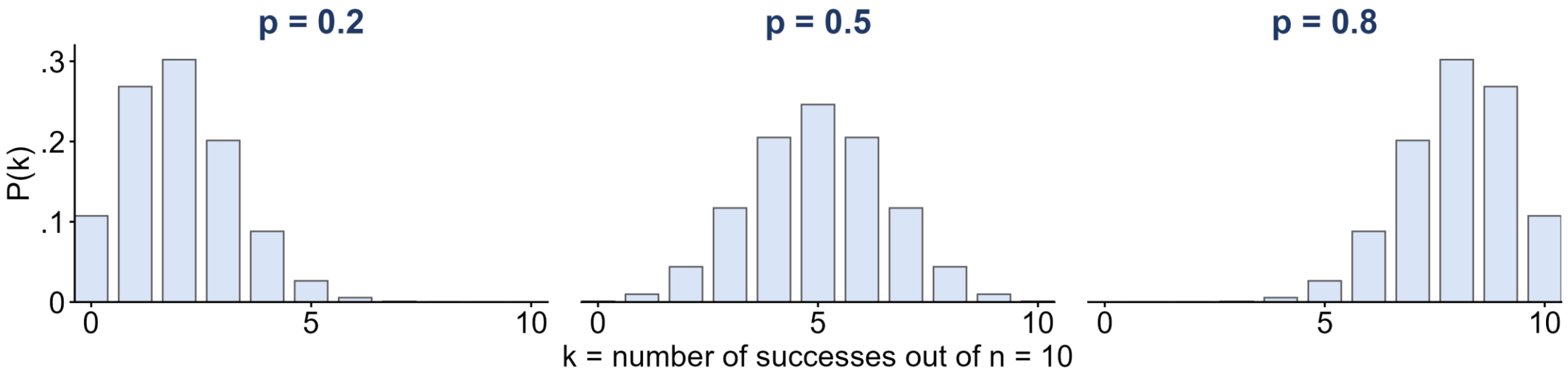
(a) What is P(X = 3)?
(b) What is P(X = 0)?
(c) What are μ and σ?

(a) $10!/(3!\,7!) \cdot 0.25^3 \cdot 0.75^7 = 120 \times 0.0156 \times 0.1335 \approx 0.250$
(b) $0.75^{10} \approx 0.056$
(c) $\mu = 10 \times 0.25 = 2.5$, $\sigma = \sqrt{(10 \times 0.25 \times 0.75)} \approx 1.37$

The shape of a binomial distribution

X ~ Binom(n = 10, p) for three values of p.

What does p do to the shape?



p = 0.5: symmetric. p < 0.5: skewed right. p > 0.5: skewed left.
The center sits at μ = np: 2, 5 and 8

Example: free throws

Stephen Curry has made 4,236 of the 4,645 free throws he has attempted in the NBA.

p = 4,236 / 4,645 ≈ 0.912; he shoots n = 10; X = free throws made

(a) What is P(he makes all 10)?
(b) What is P(he misses at least one)?
(c) How many does he make, on average?

(a) $0.912^{10} \approx 0.40$
(b) $1 - 0.40 = 0.60$
(c) $\mu = np = 10 \times 0.912 \approx 9.1$

Is it binomial?

For each X, check the three conditions: fixed n, the same p, independent trials.

(a) X = the number of heads in 20 flips of a coin
(b) X = the number of aces in a 5-card hand
(c) X = the number of rolls of a die until the first six

(a) Yes: n = 20, p = 0.5, and the flips are independent
(b) No: the cards are dealt without replacement, so the draws are not independent
(c) No: the number of trials n is not fixed

The Poisson distribution

X ~ Poisson(λ)

X = the number of events in a fixed interval of time or space, when events occur independently at an average rate λ

P(X = k)

$\lambda^k e^{-\lambda} / k!$, k = 0, 1, 2, … (e ≈ 2.718)

mean and variance

$\mu = \lambda$ and $\sigma^2 = \lambda$

Example: World Cup goals

The 2022 FIFA World Cup had 172 goals in 64 matches.

$\lambda = 172 / 64 \approx 2.69$ goals per match; X = goals in one match

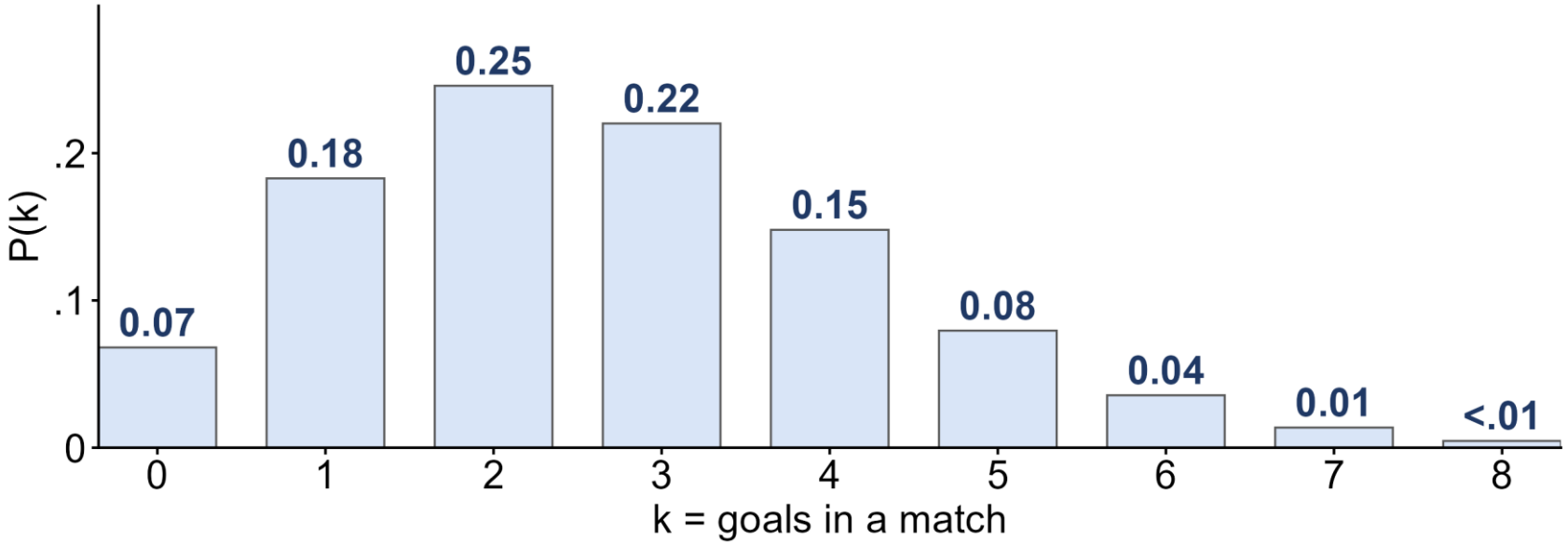
(a) What is $P(X = 0)$?
(b) How many of the 64 matches does the model expect to end 0–0? (There were 7.)
(c) What is λ for one 45-minute half, and P(no goals in a half)?

(a) $e^{-2.69} \approx 0.068$
(b) $64 \times 0.068 \approx 4.4$: close, but a model is never a perfect fit
(c) $\lambda = 2.69 / 2 \approx 1.34$, $P(X = 0) = e^{-1.34} \approx 0.26$

Goals per match: the Poisson model

X ~ Poisson(λ = 2.69): the probability of each number of goals in a match.

Which number of goals is most likely? Is the distribution symmetric?



Two goals (0.25), then three (0.22); skewed right: X cannot go below 0 but has no upper limit

Continuous random variables

density curve f(x)

the curve for a continuous X: it never goes below 0, and the total area under it is 1

probability is area

P(a < X < b) is the area under the curve between a and b; the height is not a probability

P(X = c) = 0

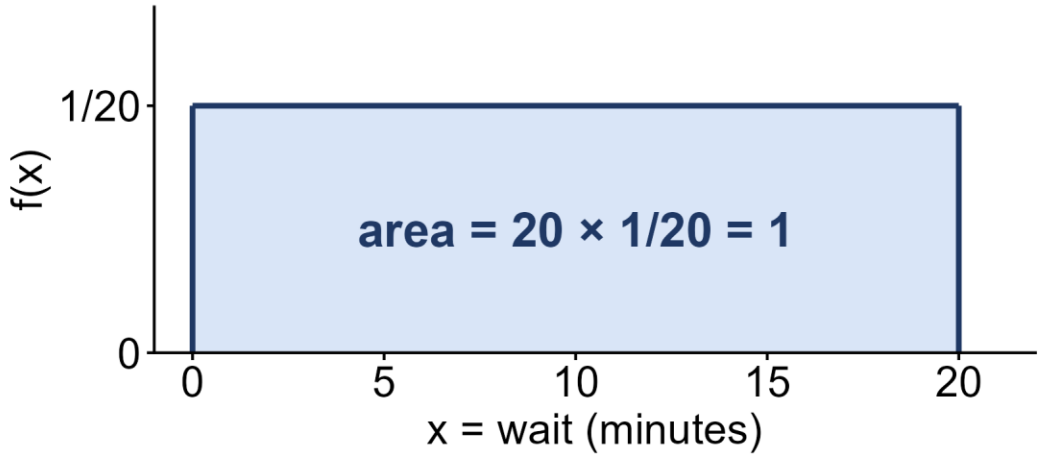
a single value has no width, so P(X < c) = P(X ≤ c)

Example: waiting for the shuttle

A campus shuttle comes every 20 minutes, and you arrive at a random time.

X = your wait in minutes: f(x) = 1/20 for $0 \le x \le 20$

(a) What is P(X < 5)?
(b) What is P(4 < X < 15)?
(c) What is P(X = 15)?

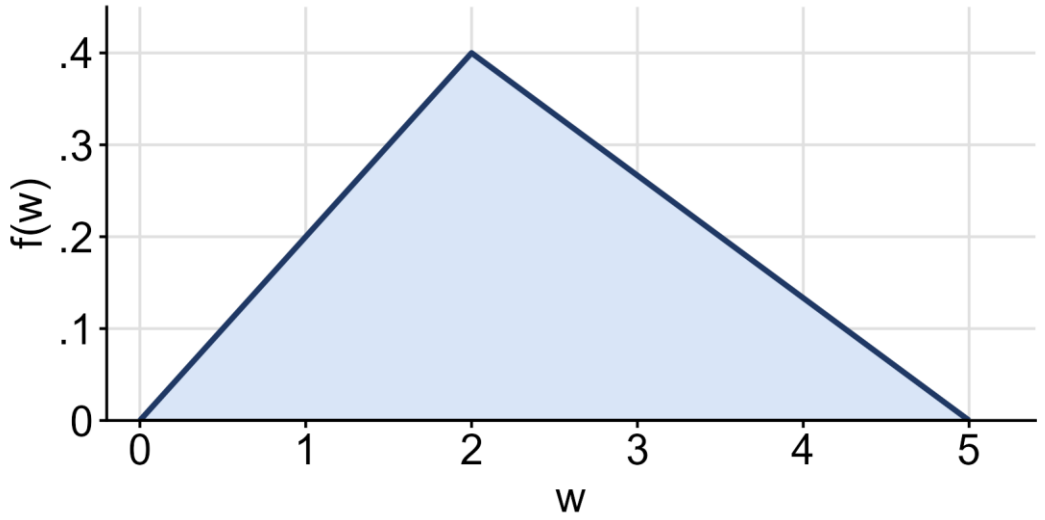


(a) 5 × 1/20 = 0.25
(b) (15 – 4) × 1/20 = 0.55
(c) 0: a single point has no width

Example: a triangle

The density of W rises in a straight line to 0.4 at w = 2, then falls to 0 at w = 5.

(a) Is the total area 1?
(b) What is P(W ≤ 1)?
(c) What is P(W > 2)?



(a) ½ × 5 × 0.4 = 1
(b) ½ × 1 × 0.2 = 0.1 (the height at w = 1 is 0.2)
(c) ½ × 3 × 0.4 = 0.6

Cumulative probability

$P(X \leq x)$

the area to the left of x: what the z-table and pnorm give

area to the right

$P(X > x) = 1 - P(X \leq x)$

area between

$P(a < X < b) = P(X \leq b) - P(X \leq a)$

Example: area between

X = the height of a woman in the student heights data, modeled as normal.

X ~ N(65.4, 3.38); Week 6: P(X < 70) = 0.9131

(a) What is P(X > 70)?
(b) What is P(X < 62)?
(c) What is P(62 < X < 70)?

(a) 1 – 0.9131 = 0.0869
(b) z = (62 – 65.4) / 3.38 = –1.01: table gives 0.1562
(c) 0.9131 – 0.1562 = 0.7569 ≈ 0.76

What you should know: distributions

expected value

$\mu = \Sigma\, x\, P(x)$: the long-run average of X

binomial

successes in n independent trials with the same p: $\mu = np$, $\sigma^2 = np(1 - p)$; Bernoulli is $n = 1$

Poisson

events in a fixed interval at an average rate λ: $\mu = \sigma^2 = \lambda$

continuous

probability is area under the density curve; $P(X = c) = 0$; $P(a < X < b) = P(X \le b) - P(X \le a)$